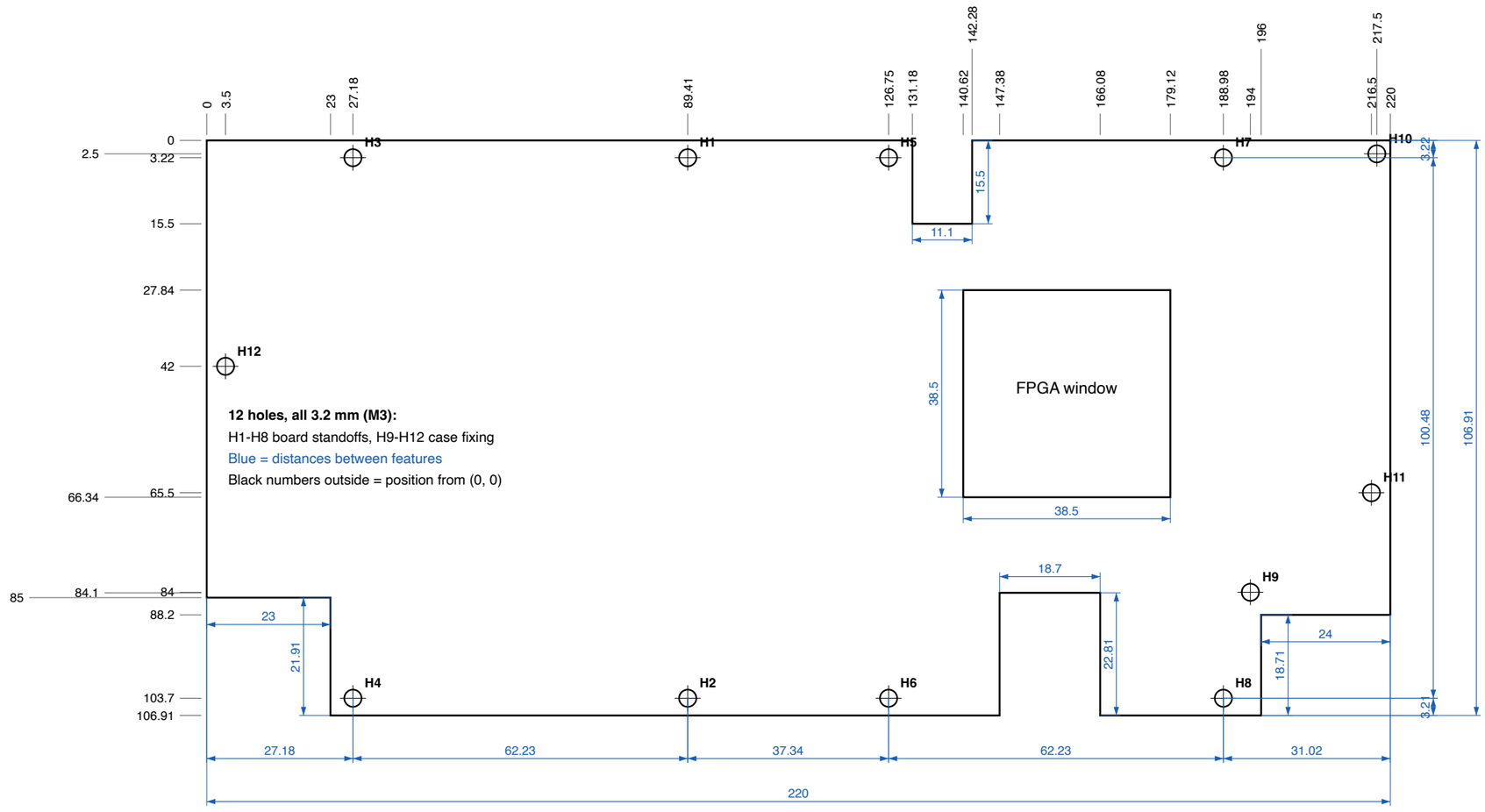
1 - Board outline, holes, notches and FPGA window

MisterFlat backplane Rev A5 - scale 1:1 on A3 - all dimensions in mm - origin (0, 0) = rear-left corner of the board, X to the right, Y towards the front



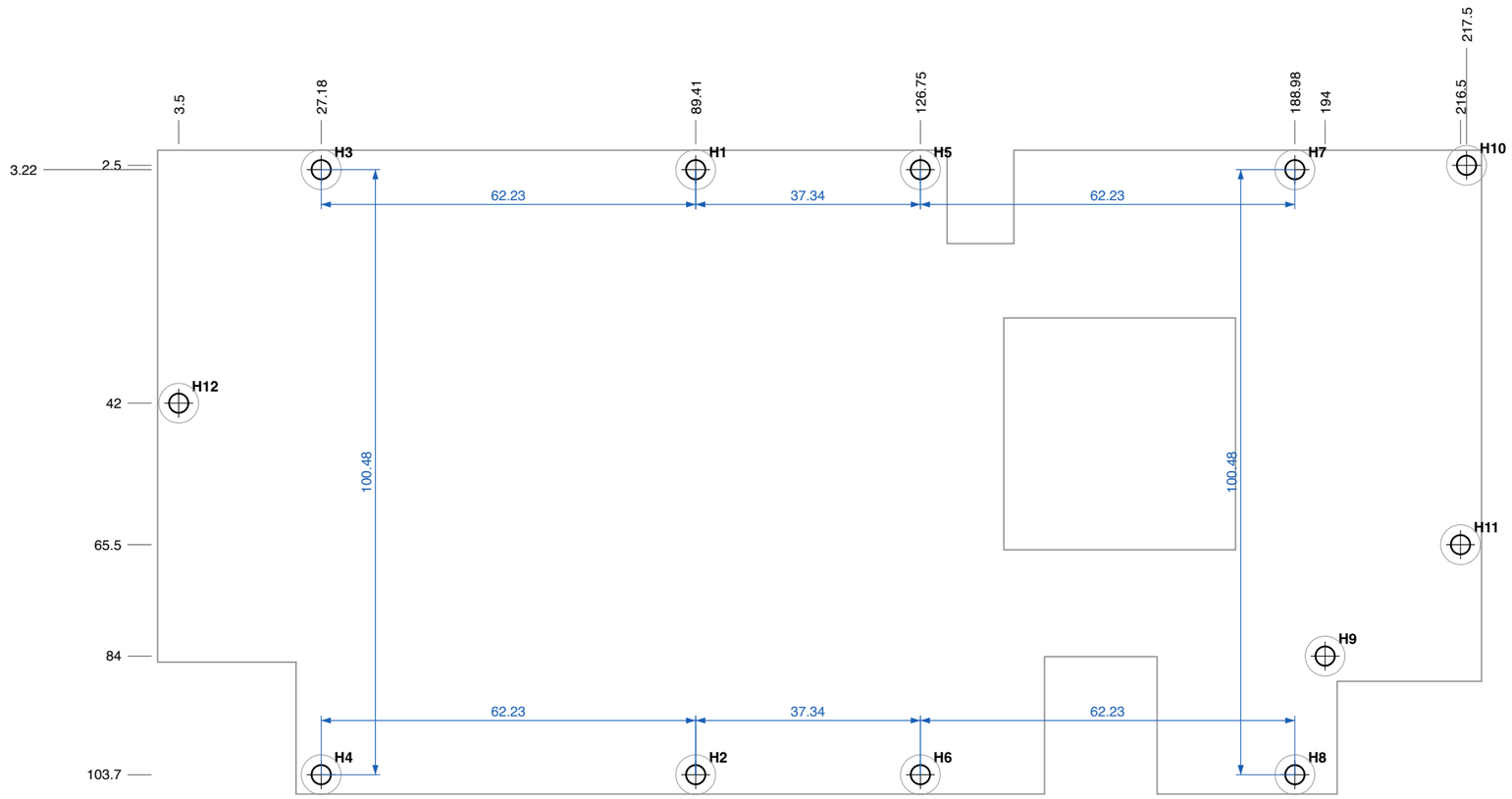
Feature	X from	X to	Y from	Y to
Board envelope	0	220	0	106.91
Power jack notch (rear)	131.18	142.28	0	15.5
Cyclone RJ45 notch (front)	147.38	166.08	84.1	106.91
Dongle notch (front-right)	196	220	88.2	106.91
Network column recess (front-left)	0	23	85	106.91
FPGA window (through hole)	140.62	179.12	27.84	66.34

Board thickness 1.6 mm. Inside corners of the cut-outs have the router radius (about 1 mm).

Hole	X	Y	Use
H1	89.41	3.22	I/O M3 standoff
H2	89.41	103.7	I/O M3 standoff
H3	27.18	3.22	I/O M3 standoff
H4	27.18	103.7	I/O M3 standoff
H5	126.75	3.22	Cyclone M3 standoff
H6	126.75	103.7	Cyclone M3 standoff
H7	188.98	3.22	Cyclone M3 standoff
H8	188.98	103.7	Cyclone M3 standoff
H9	194	84	Case M3
H10	217.5	2.5	Case M3
H11	216.5	65.5	Case M3
H12	3.5	42	Case M3

2 - Mounting holes (all 3.2 mm, M3)

MisterFlat backplane Rev A5 - scale 1:1 on A3 - all dimensions in mm - origin (0, 0) = rear-left corner of the board, X to the right, Y towards the front



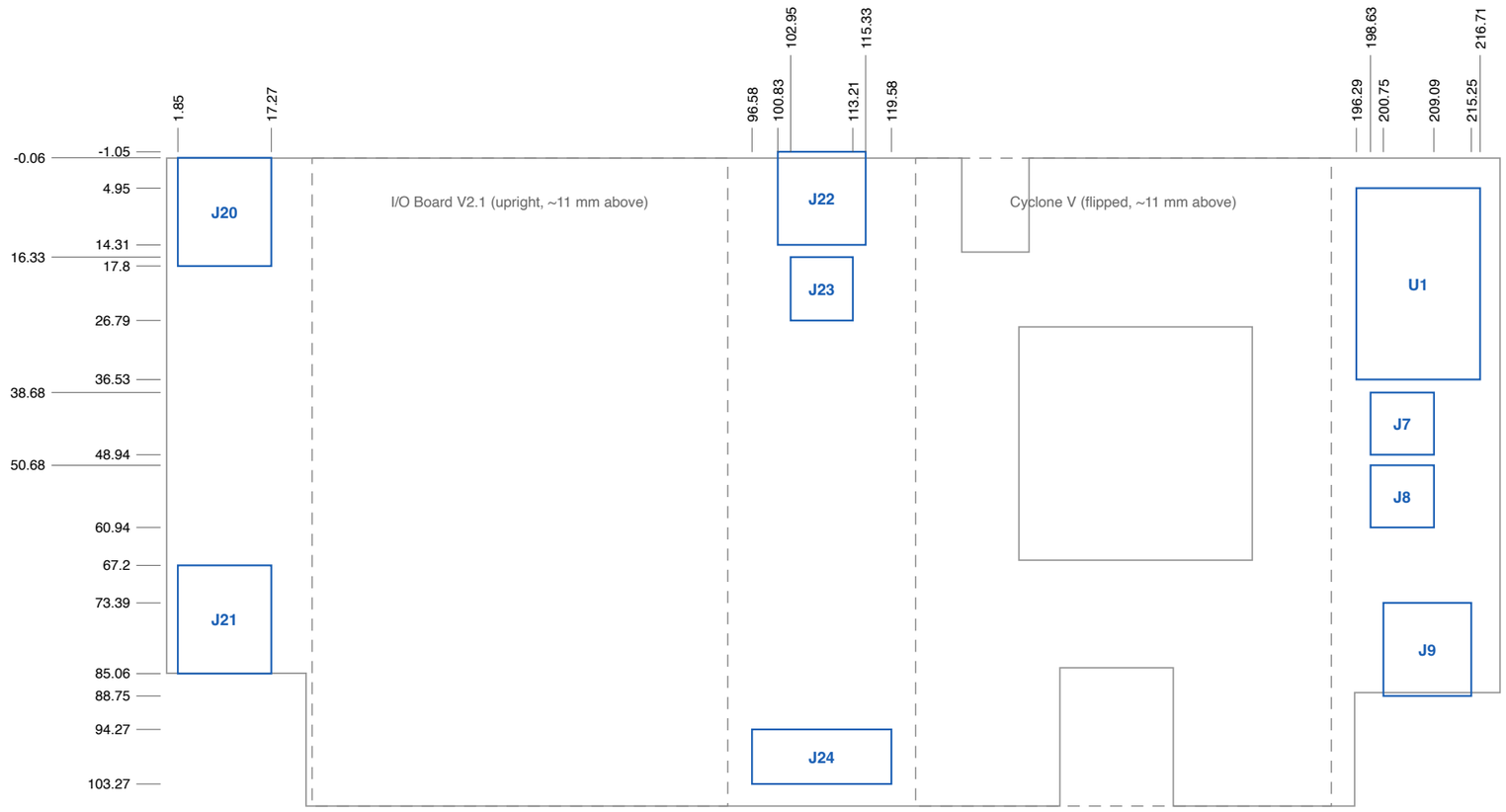
Hole	X	Y	Diameter	Use
H1	89.41	3.22	3.2	I/O M3 standoff
H2	89.41	103.7	3.2	I/O M3 standoff
H3	27.18	3.22	3.2	I/O M3 standoff
H4	27.18	103.7	3.2	I/O M3 standoff
H5	126.75	3.22	3.2	Cyclone M3 standoff
H6	126.75	103.7	3.2	Cyclone M3 standoff
H7	188.98	3.22	3.2	Cyclone M3 standoff
H8	188.98	103.7	3.2	Cyclone M3 standoff
H9	194	84	3.2	Case M3
H10	217.5	2.5	3.2	Case M3
H11	216.5	65.5	3.2	Case M3
H12	3.5	42	3.2	Case M3

Grey circle: 6.6 mm copper-free area around each hole (room for a 5.5 mm hex standoff or an M3 screw head).

H1-H4 carry the I/O Board, H5-H8 the Cyclone (about 11 mm standoffs, to measure). All of them can also fix the board to the case.

3 - Connectors on the backplane and panel openings

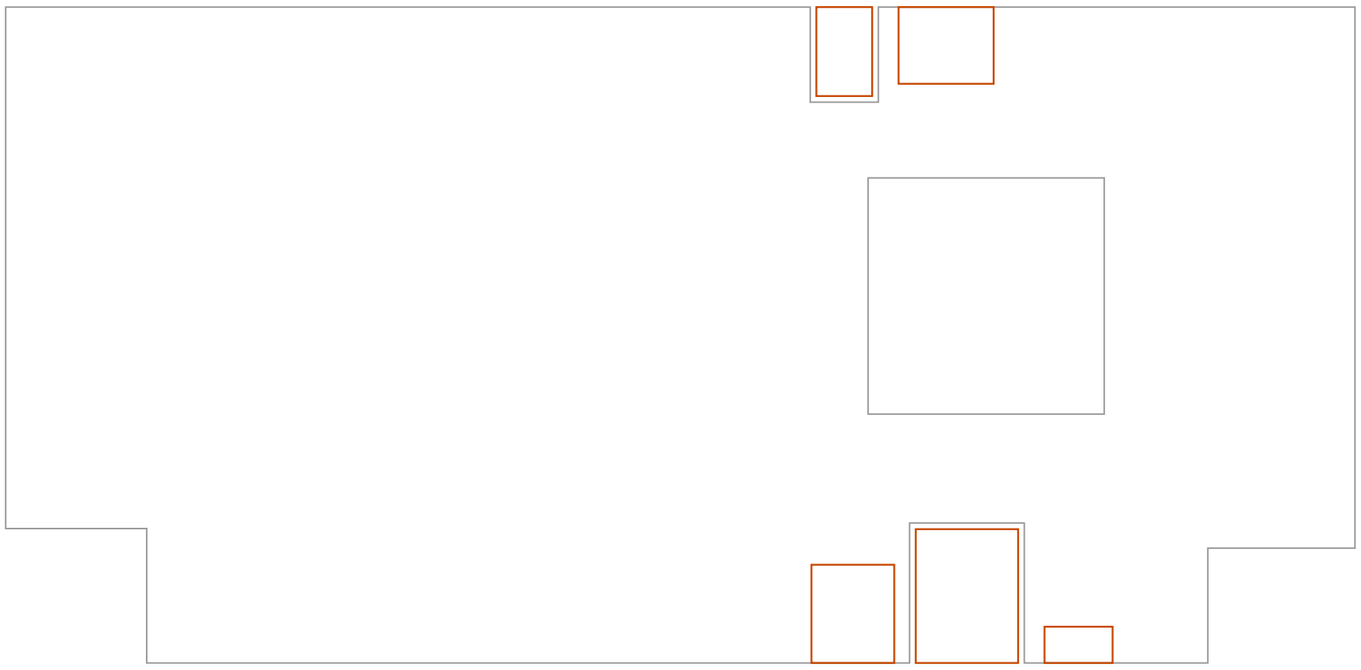
MisterFlat backplane Rev A5 - scale 1:1 on A3 - all dimensions in mm - origin (0, 0) = rear-left corner of the board, X to the right, Y towards the front



Ref	What	X from	X to	Y from	Y to	Height above PCB
J20	Network out RJ45, faces rear	1.85	17.27	-0.06	17.8	~13 (check part)
J22	Rear USB-A, faces rear	100.83	115.33	-1.05	14.31	~7
J21	Network in RJ45, faces front (recess)	1.85	17.27	67.2	85.06	~13 (check part)
J9	Wi-Fi USB-A, faces front (dongle notch)	200.75	215.25	73.39	88.75	~7
J23	Rear USB cable header, 90 deg, pins to front	102.95	113.21	16.33	26.79	~2.5 (pins) + plug
J7	AUX USB header, 90 deg, pins to right	198.63	209.09	38.68	48.94	~2.5 (pins) + plug
J8	DISPLAY USB header, 90 deg, pins to right	198.63	209.09	50.68	60.94	~2.5 (pins) + plug
J24	Buttons / LEDs header 2x6, vertical	96.58	119.58	94.27	103.27	~8.5 (pins) + plug
U1	Adafruit 5997 hub on sockets	196.29	216.71	4.95	36.53	~8.5 sockets + 1.6 + parts

4 - QMTECH board connectors at the panels and heights (estimates, measure on the real boards)

MisterFlat backplane Rev A5 - scale 1:1 on A3 - all dimensions in mm - origin (0, 0) = rear-left corner of the board, X to the right, Y towards the front



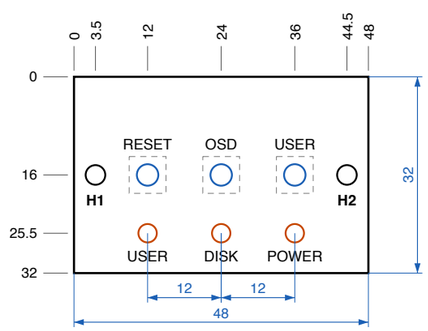
Connector (on the QMTECH board)	X from	X to	Y from	Y to
Cyclone HDMI (rear)	145.58	161.08	0	12.5
Cyclone power jack (rear)	132.18	141.28	0	14.5
Cyclone RJ45 (front)	148.38	165.08	85.1	106.91
Cyclone USB-A J13 (front, unused)	131.38	144.88	90.9	106.91
Cyclone mini-USB UART J4 (front)	169.38	180.48	101	106.91

Height stack (Z, from the backplane top face)	mm
Backplane PCB thickness	1.6
Cyclone component face (flipped board) above backplane	~11 (socket 8.5 + header 2.5), measure
Cyclone PCB top (its back side) above backplane	~12.6 + its bottom parts (~2)
I/O Board PCB bottom above backplane	~11, measure
I/O Board parts on top	unknown, measure
Heatsink / fan below the backplane (through the window)	unknown, measure
Standoffs H1-H8	= mated connector height (~11)

Orange boxes: QMTECH board connector bodies from the QMTECH drawing (raster estimate, +-0.5 mm). The I/O Board V2.1 rear connectors (VGA, audio) are not known yet.

5 - Front panel board (buttons and LEDs) - drilling template for the case front

MisterFlat front panel Rev A - scale 1:1 on A3 - all dimensions in mm - seen from the front (button side) - origin (0, 0) = top-left corner of the board



Feature	X	Y	Size	Case front hole
Board	0 .. 48	0 .. 32	1.6 thick	
H1 mounting hole	3.5	16	3.2 (M3)	M3 standoff or boss
H2 mounting hole	44.5	16	3.2 (M3)	M3 standoff or boss
Button RESET	12	16	6x6 body, 3.5 stem	4.5-5 for the stem
Button OSD	24	16	6x6 body, 3.5 stem	4.5-5 for the stem
Button USER	36	16	6x6 body, 3.5 stem	4.5-5 for the stem
LED USER	12	25.5	3 mm LED	3.2, or 5 with a clip
LED DISK	24	25.5	3 mm LED	3.2, or 5 with a clip
LED POWER	36	25.5	3 mm LED	3.2, or 5 with a clip

Blue circle: button stem (13 mm above the PCB). Orange circle: LED body. Dashed: 6x6 button body.
The board stands upright behind the case front wall, buttons and LEDs facing out.
J1 (IDC 2x6) is on the back side, at the top: leave about 12 mm behind the board for the ribbon plug.
Board-to-wall distance: about 13 mm minus the wall thickness, so the stems come out about 1 mm.
Solder the LEDs with long legs so they reach the wall, or use a light pipe.